

# Customer Experiences: Digital Transformation, Downtime-Optimized RISE Migrations & AI in Action

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**01** • Digital Transformations

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**02** • Customer Exp – doDMO

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**03** • Customer Exp – ZDO

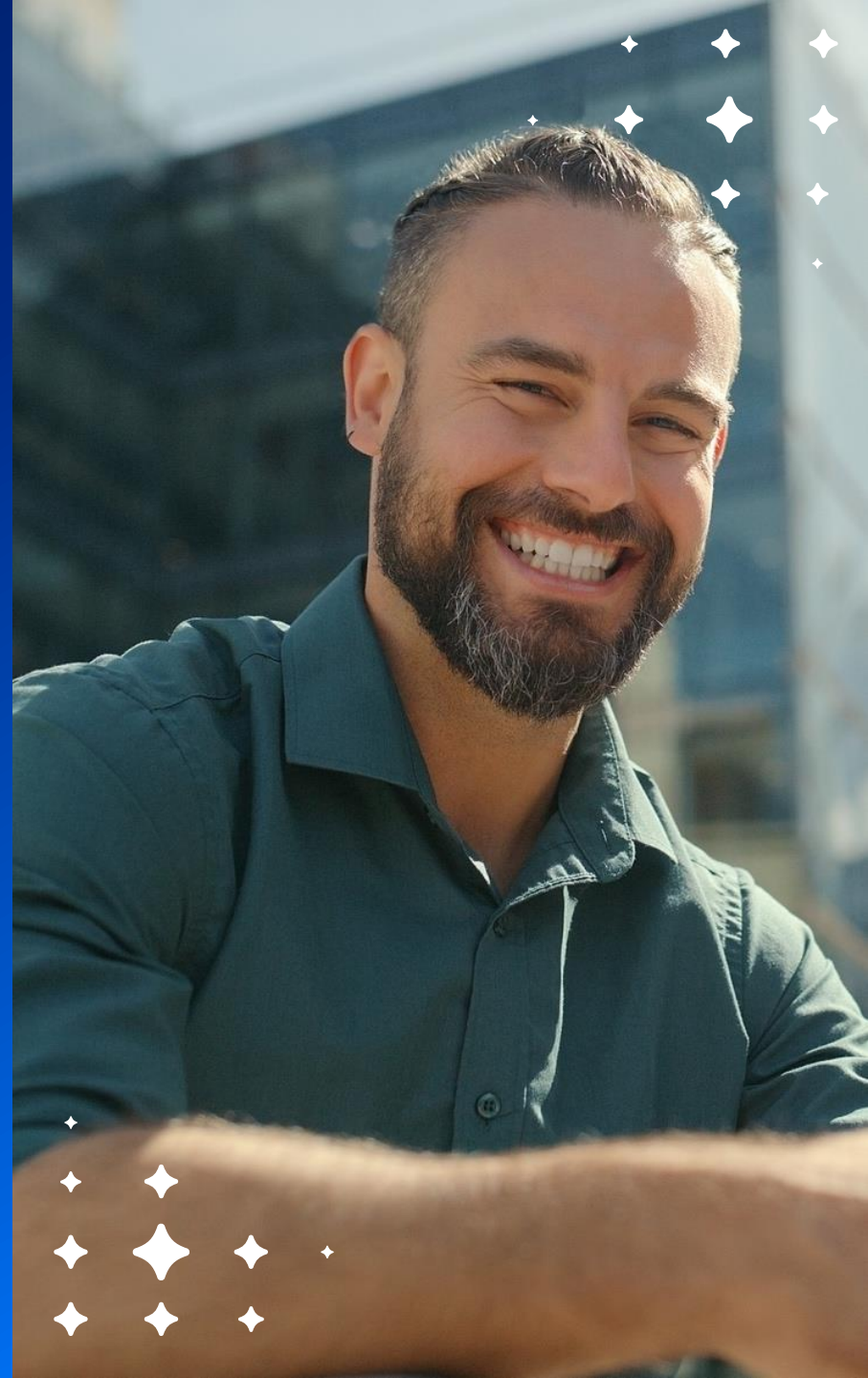
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**04** • Customer Exp – doC (Cust on TEMP)

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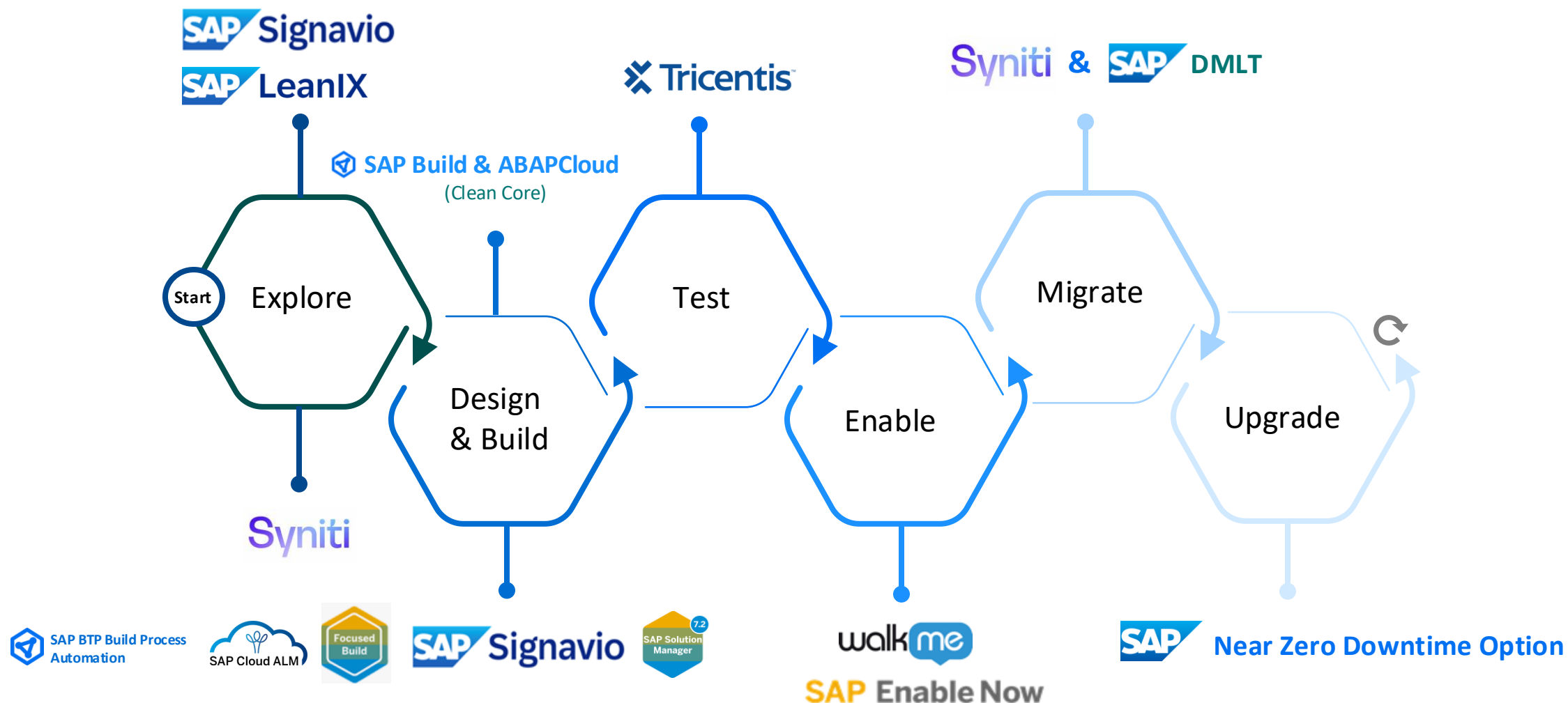
**05** • AI in Action

# Digital Transformation



# The Big Picture: SAP Tools supporting an accelerated S/4HANA Roll-out

## Systematically bridging Business and IT Transformation

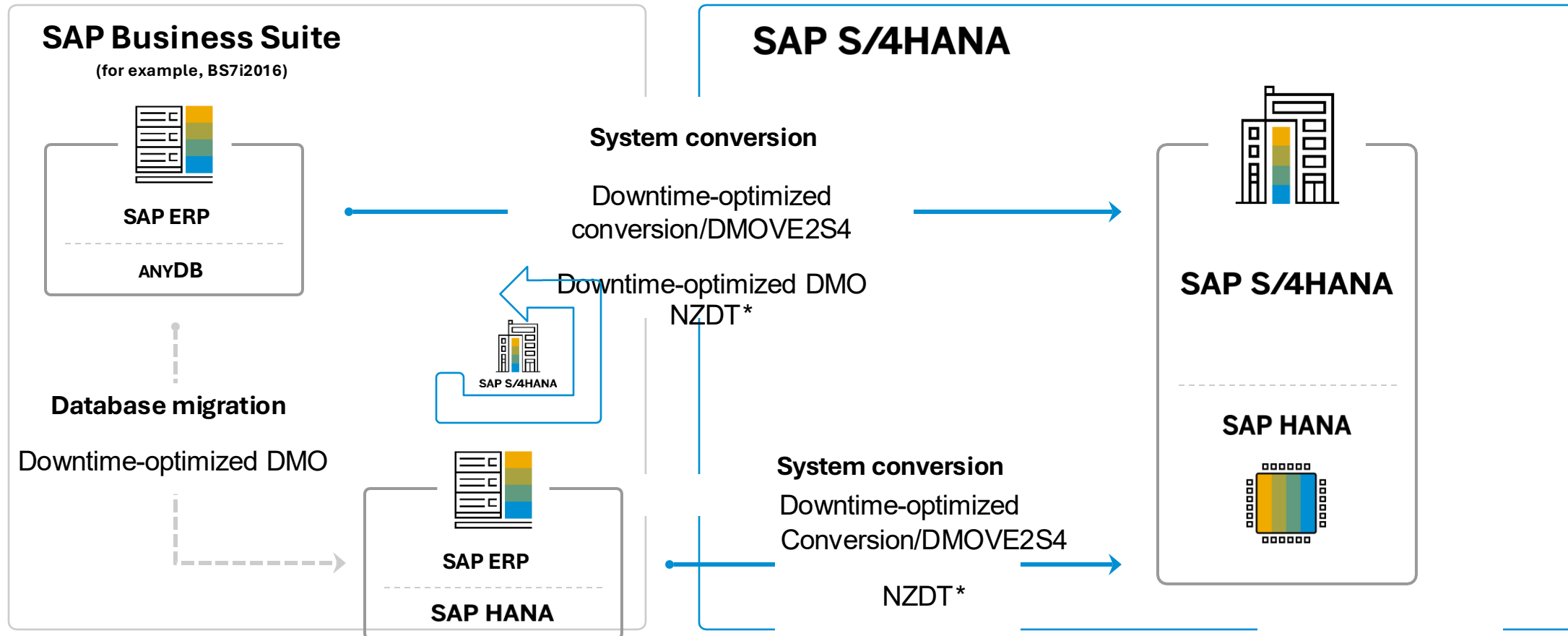


# Customer Experience – Downtime Optimized Maintenance





# Downtime optimization approaches for system maintenance



\* NZDT is service-based and currently being redesigned

# Downtime-optimization approaches for maintenance activities

| Approach                                     | Abbreviation                  | Scenario                                  | Availability*                   | SAP Note                                                                        |
|----------------------------------------------|-------------------------------|-------------------------------------------|---------------------------------|---------------------------------------------------------------------------------|
| near-Zero Downtime Maintenance (ABAP) **     | <b>nZDM</b> (ABAP)            | Update/Upgrade                            | Unrestricted available          | <a href="#">1678565</a>                                                         |
| Zero Downtime Option                         | <b>ZDO</b>                    | Update/Upgrade                            | Pilot<br>Unrestricted available | Business Suite: <a href="#">2163060</a><br>SAP S/4HANA: <a href="#">2707731</a> |
| downtime-optimized Database Migration Option | <b>downtime-optimized DMO</b> | Migration to SAP HANA database            | Unrestricted available          | <a href="#">2547309</a>                                                         |
| downtime-optimized Conversion                | <b>doC</b>                    | Conversion to SAP S/4HANA                 | Available for educated experts  | <a href="#">doC Portal Page</a>                                                 |
| Near Zero Downtime Technology                | <b>NZDT</b>                   | Multiple (such as SAP S/4HANA Conversion) | Service-based                   | <a href="#">693168</a>                                                          |

\* This is the current state of planning and may be changed by SAP at any time.

\*\* This is different to nZDM for SAP HANA database

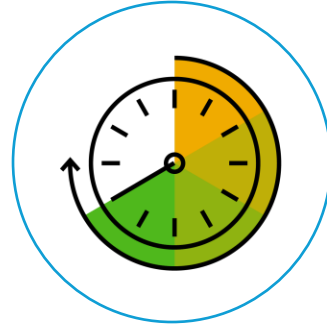


# SAP S/4HANA conversion methodologies



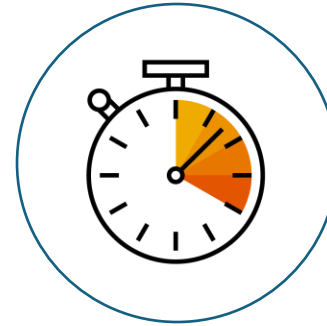
- **Standard approach**

- General availability
- SAP Note [3106931](#)



- **downtime-optimized Database Migration Option**

- General availability
- SAP Note [2547309](#)



- **downtime-optimized Conversion**

- General availability
- [doC Portal Page](#)



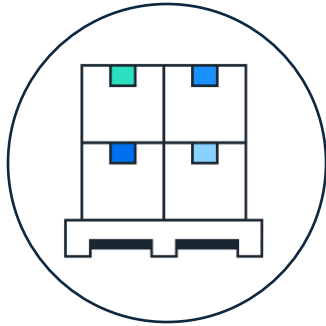
- **Near-Zero Downtime Technology**

- Minimized Downtime Service
- SAP Note [693168](#)

EFFORT

DOWNTIME

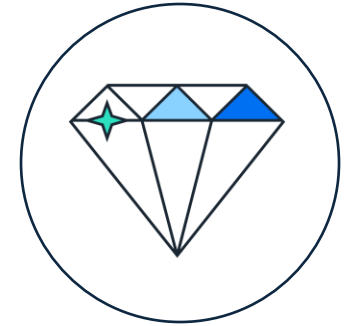
# Options for updating SAP S/4HANA using the Software Update Manager (SUM)



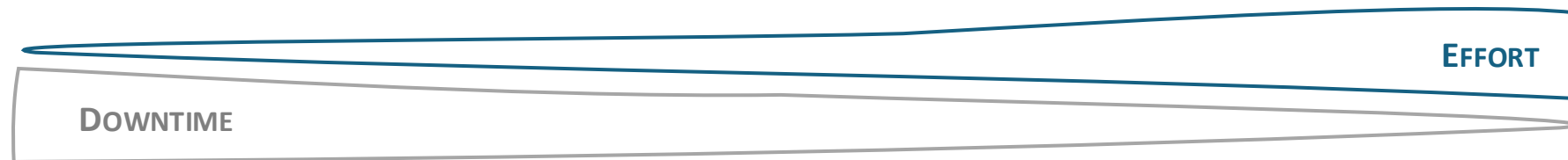
- Standard approach
- Several downtime optimizations (for example, shadow system operations)



- Near-Zero Downtime Maintenance (nZDM)
- Downtime reduction by moving downtime-relevant phases to uptime

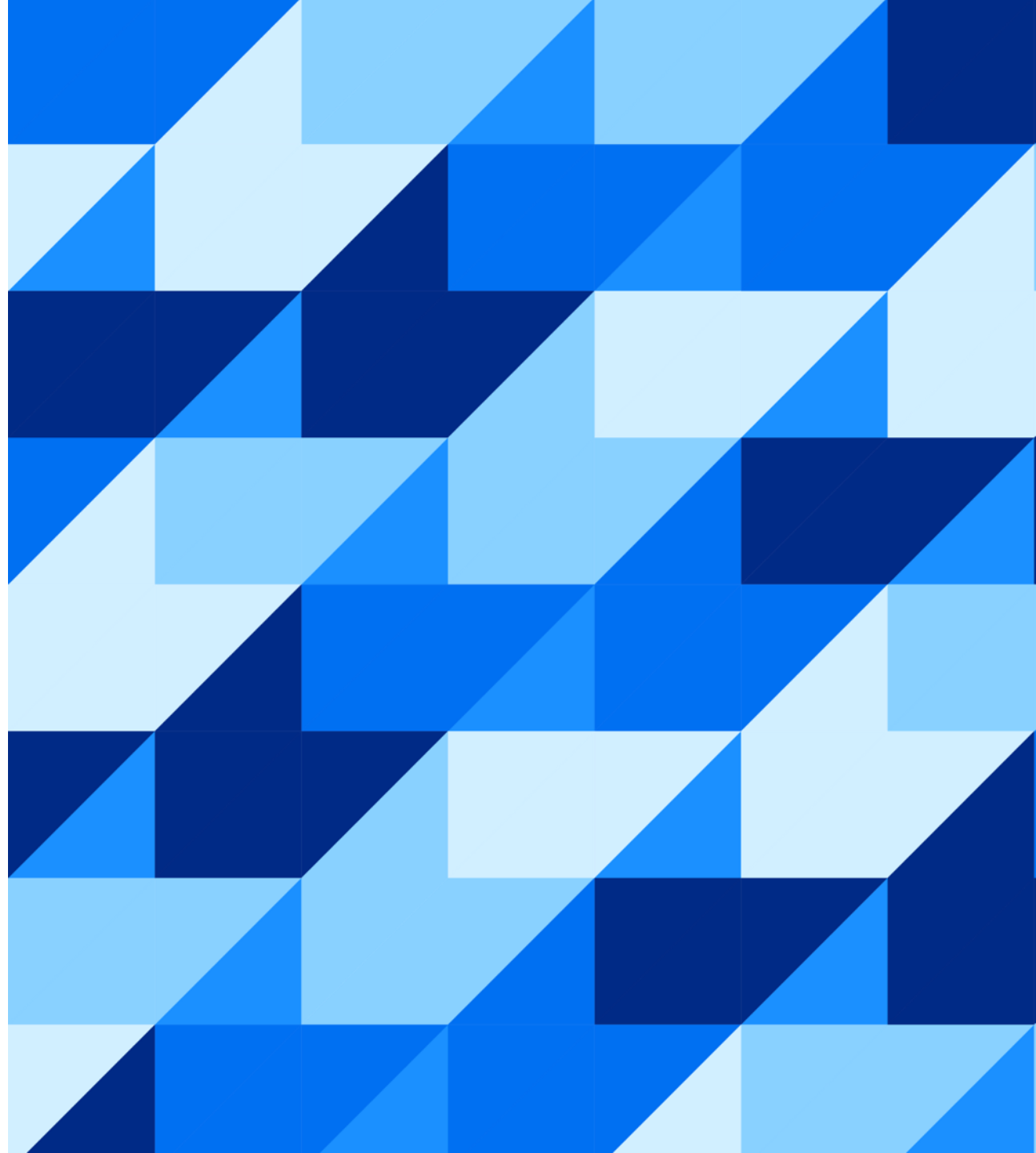


- Zero Downtime Option (ZDO)
- Complete upgrade running in parallel with continuous business operations



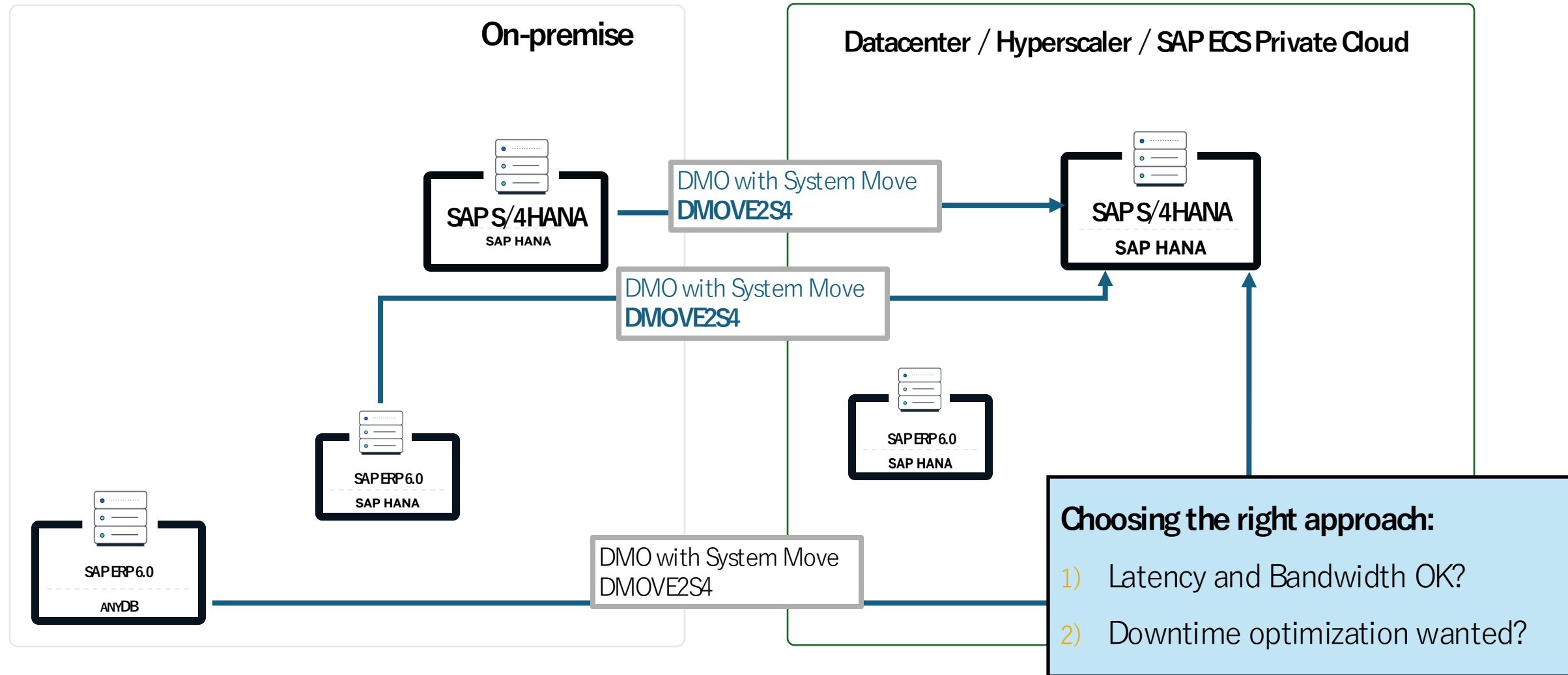
# Customer Success Stories – DMOVE2S4 & doDMO

INTERNAL – SAP and Customers Only



# Transition Approaches and Tools

## “DMO with System Move” and “DMOVE2S4”

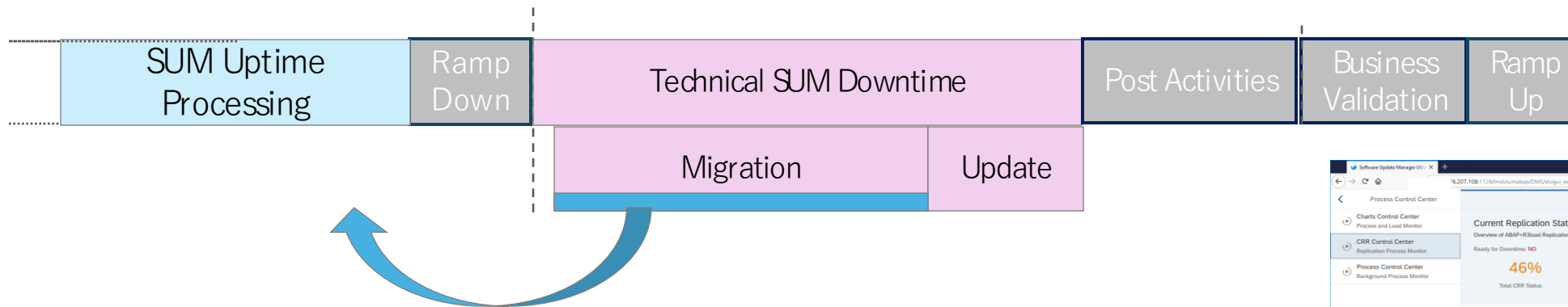


Recommendation for DMOVE2S4: latency lower than 20ms and bandwidth higher than 400 Mbps. See SAP Note [3620006](#)

# Transition Approaches and Tools

Downtime-optimized Approach – **downtime-optimized DMO**

- Generally available with SUM 2.0 SP 06 (and higher) for system conversions and migrations
- **Uptime migration** for selected large application tables
  - User changes are reflected with record-and-replay technology of SUM
  - Includes initial and delta migration in uptime, remaining delta migration in downtime



- Option is offered on SUM dialog on scenario strategy
- Report available to select appropriate tables for uptime migration
- Replication monitoring is part of SUM Utilities
- Applicable only if source is not yet on SAP HANA database

The screenshot shows the 'Current Replication Status' in the SUM Utilities interface. It displays the following information:

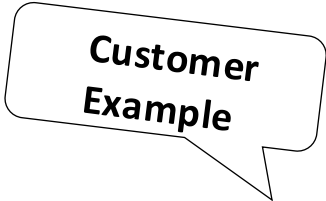
- Ready for Downtime:** NO
- Total CRR Status:** 46%
- ABAP CRR Status:** 47%
- R3load CRR Status:** 0%

Below this, there are sections for 'ABAP Replication Monitor' and 'R3load Replication Monitor', both showing 'running' status. At the bottom, there is a table with the following columns: Table Name, Replay Status, Recorded Changes, Replicated Changes, Pending Changes, and Percentage (in %).

| Table Name | Replay Status | Recorded Changes | Replicated Changes | Pending Changes | Percentage (in %) |
|------------|---------------|------------------|--------------------|-----------------|-------------------|
| EDK40      |               | 0                | 0                  | 0               | 0                 |
| CDCLS      |               | 0                | 0                  | 0               | 0                 |

<https://community.sap.com/t5/technology-blogs-by-sap/dmo-downtime-optimization-by-migrating-app-tables-during-uptime/ba-p/13108040>

S/4HANA 2023 Brownfield conversion | Live  
Source 9 TB | 60 hours business downtime



Customer

- FMCG Industry
- Single Global S/4 HANA system

Project

- S/4 Conversion and RISE Migration
- Duration 9 months
- **Source DB Size: 8.7 TB**
- **Target DB size: 2 TB**

Technical Scope

- Source SAP Version : ECC 6.0 EHP6
- Target SAP Version : S/4 HANA 2023 FPS03
- SUM Version : SUM 2.0 SP25
- **Technical Downtime : ~11 hours using DoDMO**
- SAP landscape on RISE

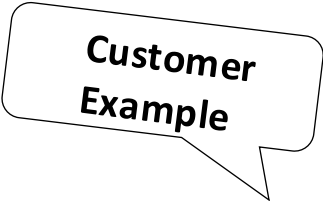
Scope

- **S/4HANA Migration Scope:** Migrate ERP system to S/4HANA 2023 FPS03 with minimum changes approach
- **Fiori:** Embedded Fiori Activation

Key Figures

- **80% reduction in downtime with Downtime Optimized DMO approach**
- 3 Mocks of production copy were performed
- Migration time was brought down from 50 hours to 1.5 hours.

S/4HANA 2023 Brownfield conversion | Live  
Source 18 TB | 60 hours business downtime



Customer

- Utility Industry
- Single Global S/4 HANA system
- Multiple parallel Tracks

Project

- S/4 Conversion and RISE Migration
- Duration 12 months
- **Source DB Size: 18 TB(Compressed)**
- **Target DB size: 7 TB**

Technical Scope

- Source SAP Version : ECC 6.0 EHP 8
- Target SAP Version : S/4 HANA 2023 FPS03
- SUM Version : SUM 2.0 SP25
- **Technical Downtime : ~14 hours using DoDMO**
- SAP landscape on RISE

Scope

- **S/4HANA Migration Scope:** RISE Migration and S/4 HANA Conversion
- **Fiori:** Embedded Fiori Activation
- Build Work Zone Activation
- Joule Activation
- BW/4 HANA Green Field Implementation

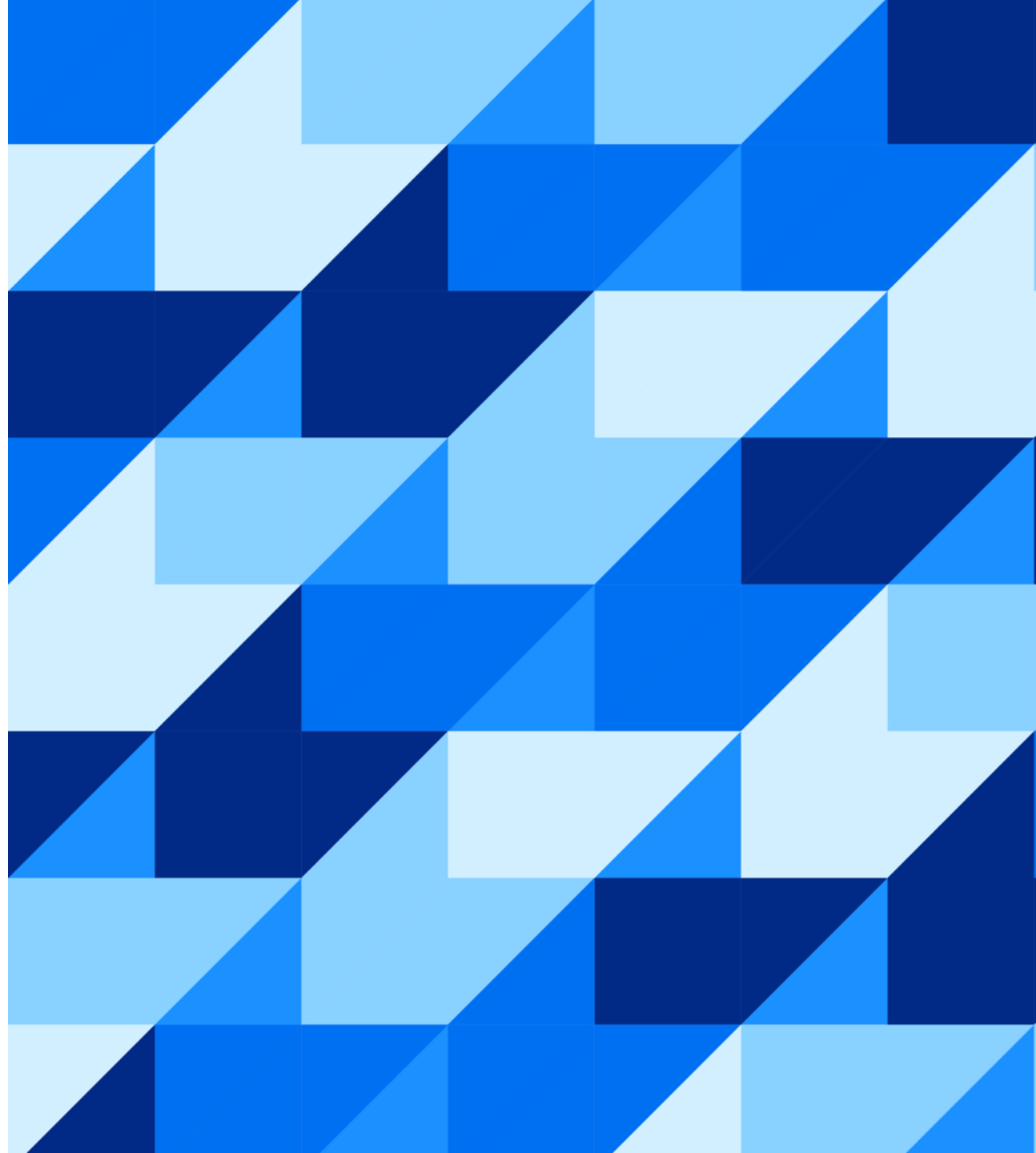
Key Figures

- **90% reduction in downtime with Downtime Optimized DMO approach**
- 3 Mocks of production copy were performed
- Migration time was brought down from 70 hours to 4 hours.



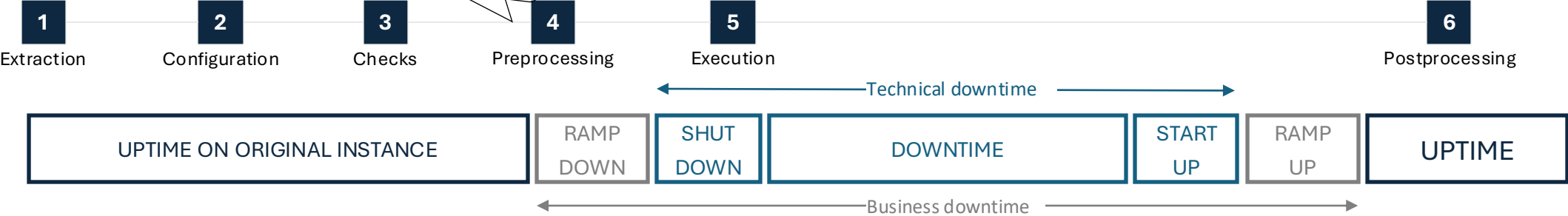
# Customer Success Stories – ZDO

INTERNAL – SAP and Customers Only

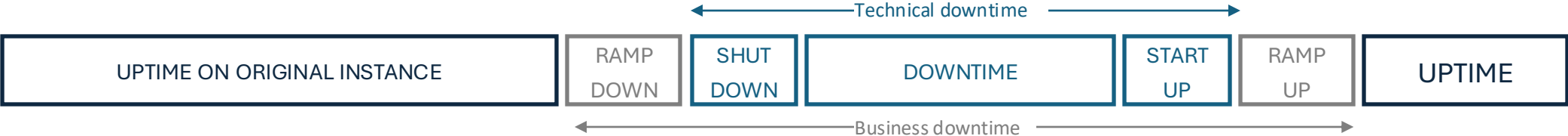


# Downtimes of the different upgrade approaches

Workbench Lock



- Standard approach



- Near-Zero Downtime Maintenance (nZDM)



- Zero Downtime Option (ZDO)



## Mahindra & Mahindra seamlessly performs upgrades with **Zero Downtime** Option of Software Update Manager

### Challenge

Mahindra & Mahindra Limited operates a colossal 2 x 9TB scale-out SAP S/4HANA Private Cloud production system. To stay up-to-date, Mahindra planned to upgrade to the current major release. However, minimizing business disruption is a priority, especially in the fast-paced automotive manufacturing industry where even the slightest downtimes can have critical impact.

### Solution

To minimize unavailability of the system, Mahindra leveraged the Zero Downtime Option (ZDO) of Software Update Manager. This downtime-optimization approach is offered with an additional service under the SAP Enterprise Cloud Services (ECS). Through close collaboration between SAP MaxAttention, Enterprise Cloud Services Upgrade Excellence Center and Cloud Lifecycle Management development teams, Mahindra completed a major release upgrade – from SAP S/4HANA 2020 to SAP S/4HANA 2023 – with a system unavailability of only 20 minutes.

### Major achievements

- Overall business downtime of 20 hours
- Minimal disruption for Mahindra's cloud experience
- Successful ZDO handling for massive scale-out system (18TB in total)

"By leveraging ZDO, we significantly reduced the downtime compared to our previous upgrades, which typically required a 48-hour window. The project's success can be attributed to the exceptional collaboration among the Mahindra & Mahindra and SAP teams."

Sanjay Chavan, VP & Head of Enterprise Applications, Mahindra & Mahindra

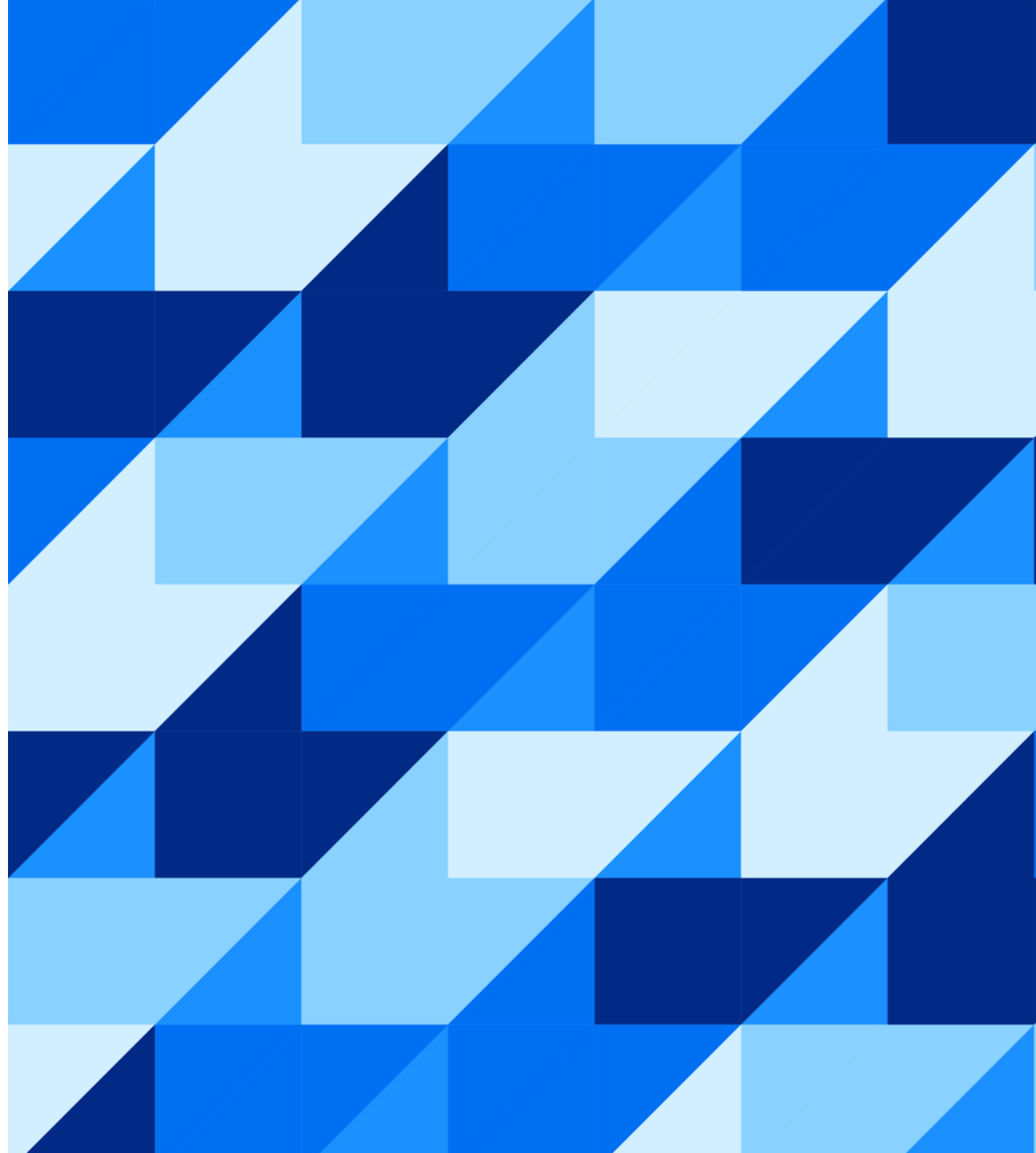
- **20 mins**
  - Unavailability of the system for a major release upgrade

- **18 TB**
  - Size of the S/4HANA production system that was successfully upgraded using ZDO

|                                                                     |              |                                |             |                  |                                                 |
|---------------------------------------------------------------------|--------------|--------------------------------|-------------|------------------|-------------------------------------------------|
| • Company Name                                                      | • Industry   | • Product and Services         | • Employees | • Revenue        | • Featured solutions                            |
| • Mahindra & Mahindra Limited                                       | • Automotive | • Passenger vehicles           | • >140.500  | • US\$16 billion | Zero Downtime Option of Software Update Manager |
| • <a href="https://www.mahindra.com/">https://www.mahindra.com/</a> |              | • Military vehicles            |             |                  |                                                 |
|                                                                     |              | • Digital Farming Technologies |             |                  |                                                 |

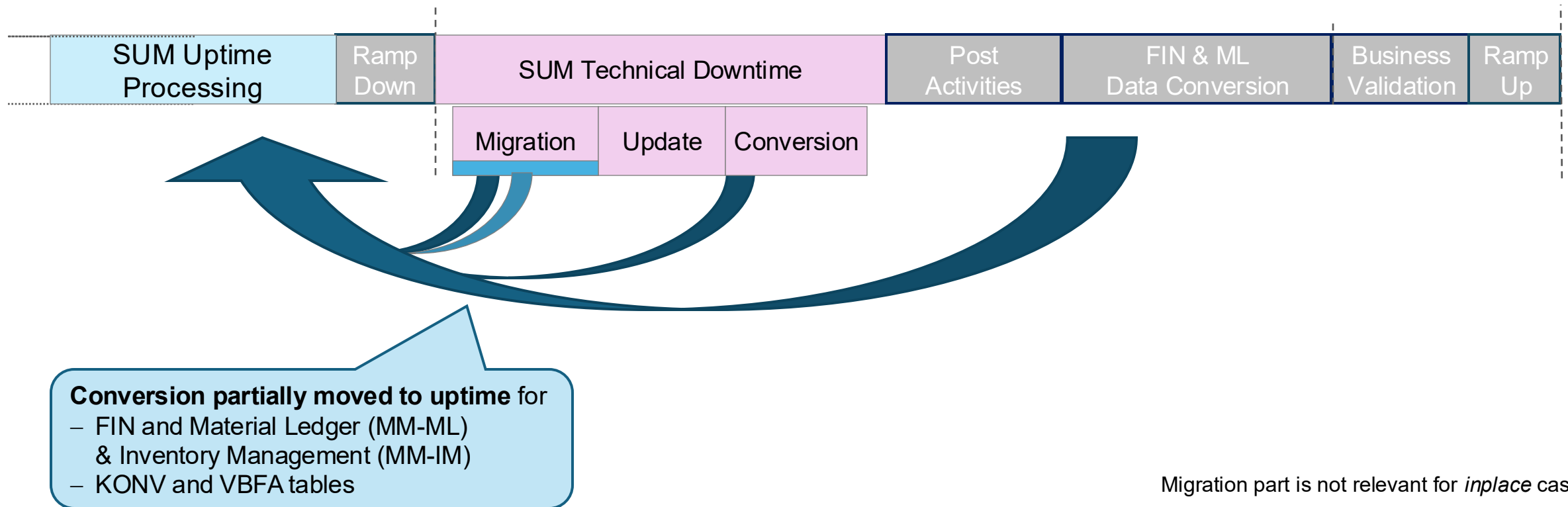
# Customer Success Stories – doC (CustOnTemp)

INTERNAL – SAP and Customers Only



# Downtime-optimized conversion approach

- **Data conversion** moved to uptime processing, also for FIN & ML
- **Field conversion** moved to uptime processing (KONV and VBFA tables)
- **Uptime migration** for selected large application tables (which are not part of the data conversion)

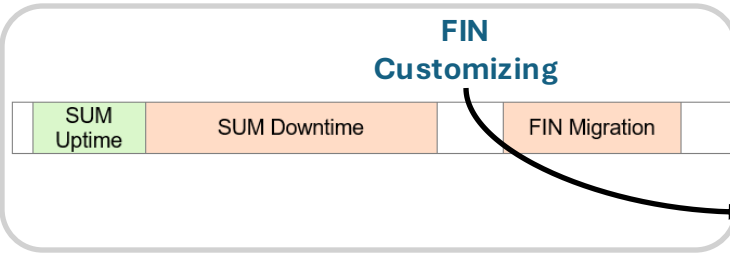


Migration part is not relevant for *inplace* case

# Downtime-Optimized Conversion with CustOnTmp

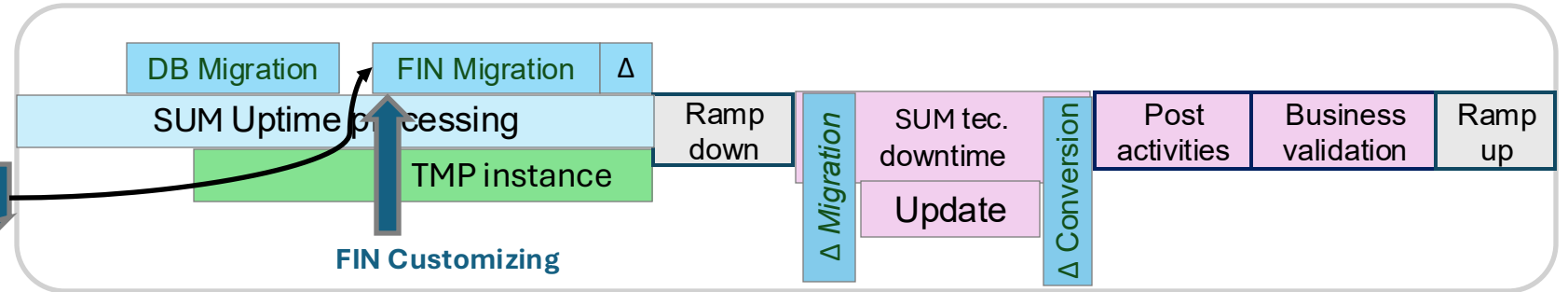
“Customizing on Temporary Instance” in a Nutshell

## Standard Conversion



Customizing Request

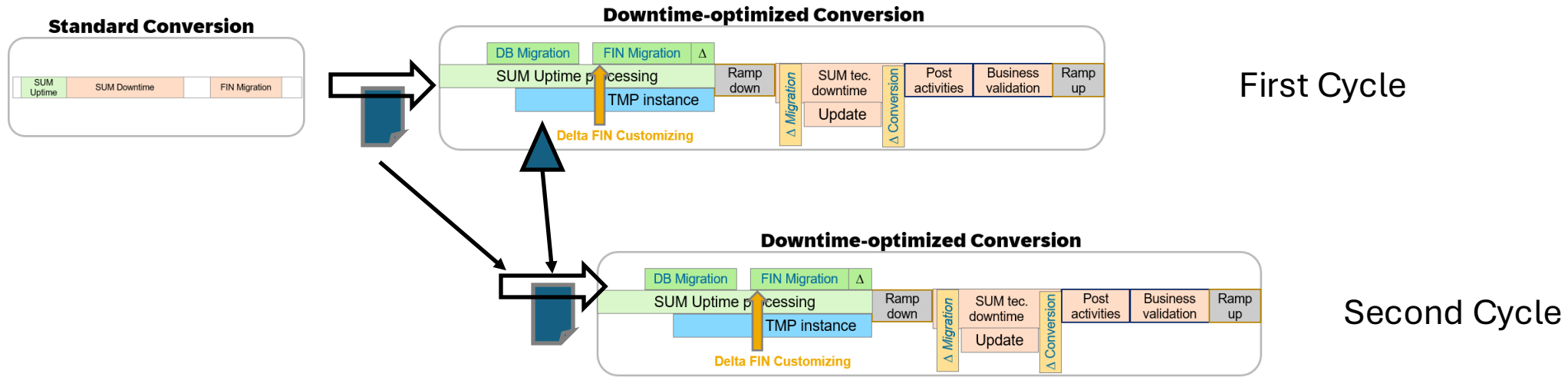
## Downtime-optimized Conversion



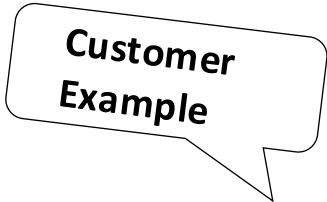
- The downtime-optimized Conversion approach executes **FIN data migration** already in uptime
- This requires the **FIN customizing** (on S/4HANA-level) to be provided beforehand, created by a standard conversion run prior to the downtime-optimized conversion run
- This requirement of „always a standard conversion run beforehand“ is eliminated by *CustOnTmp*: the FIN customizing can be adapted **during** the downtime-optimized conversion run
- SUM always creates a second instance (here: temporary instance, TMP) on target level
- This TMP instance is extended to enable customizing on target level, parallel to original instance

# Downtime-Optimized Conversion with CustOnTmp

## Simplified Procedure



- One standard conversion run will be done to create the initial FIN customizing
- All subsequent downtime-optimized conversion runs consume this FIN customizing
- Any FIN customizing changes in PRD after the standard run are reflected with delta customizing activity on the TMP instance



Customer

- Entertainment Industry
- Single Global S/4 HANA system

Project

- S/4 Conversion
- Duration 29 months
- Source DB size: 38TB
- Target DB Size : 11.7TB

Technical Scope

- Source SAP Version : ECC 6.0 EHP7
- Target SAP Version : S/4 HANA 2022 FPS03
- SUM Version : SUM 2.0 SP19
- Technical Downtime : 15 hours using DoC
- SAP landscape on premise

Scope

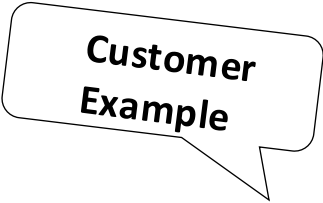
- **S/4HANA Migration Scope:** Convert ERP system to S/4HANA 2022 with minimum changes approach
- **Fiori:** Embedded Fiori Activation

Key Figures

- **85% reduction in the downtime with DoC-CustOnTmp approach**
- 7 Mocks of production copy were performed
- 2.6 Billion delta records replicated in uptime.



S/4HANA 2023 Brownfield conversion | Live  
Source 17 TB | 60 hours business downtime



Customer

- FMCG Industry
- Single Global S/4 HANA system

Project

- S/4 Conversion
- Duration 18 months
- Source DB Size: 17TB
- Target DB size: 4.5TB

Technical Scope

- Source SAP Version : ECC 6.0 EHP7
- Target SAP Version : S/4 HANA 2023 FPS01
- SUM Version : SUM 2.0 SP22
- Technical Downtime : ~8 hours using DoC
- SAP landscape on premise

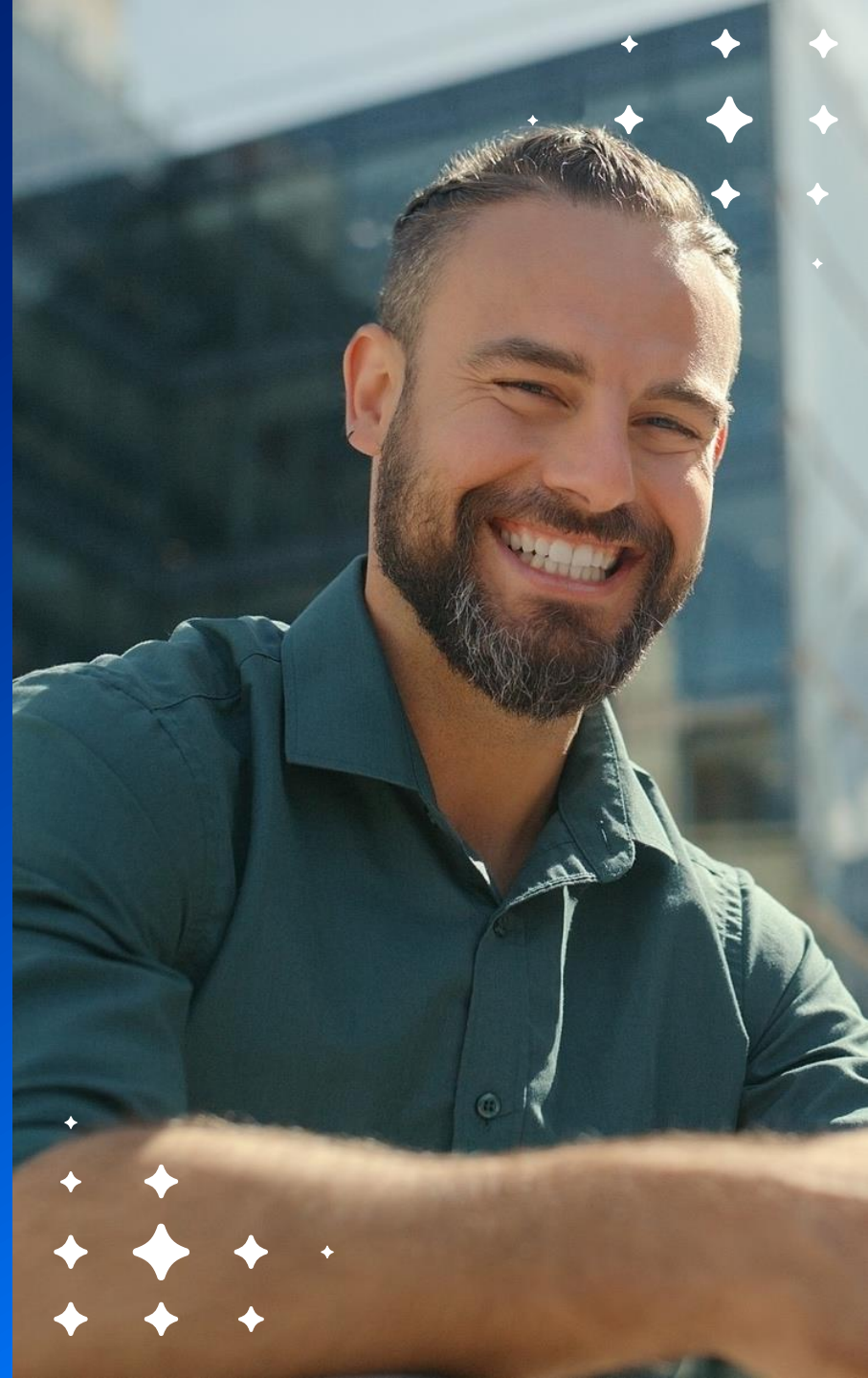
Scope

- **S/4HANA Migration Scope:** Migrate ERP system to S/4HANA 2023 with minimum changes approach
- **Fiori:** Embedded Fiori Activation

Key Figures

- **85% reduction in downtime with DoC – CustOnTmp approach**
- 7 Mocks of production copy were performed
- ~500 million delta records replicated in uptime

# AI in Action



# Ambition migration and modernization



Approximately

# 35%

reduction of migration effort with an Agent-led toolchain – and a path to mostly autonomous transitions

## Step 1

### 2026: AI-supported migration

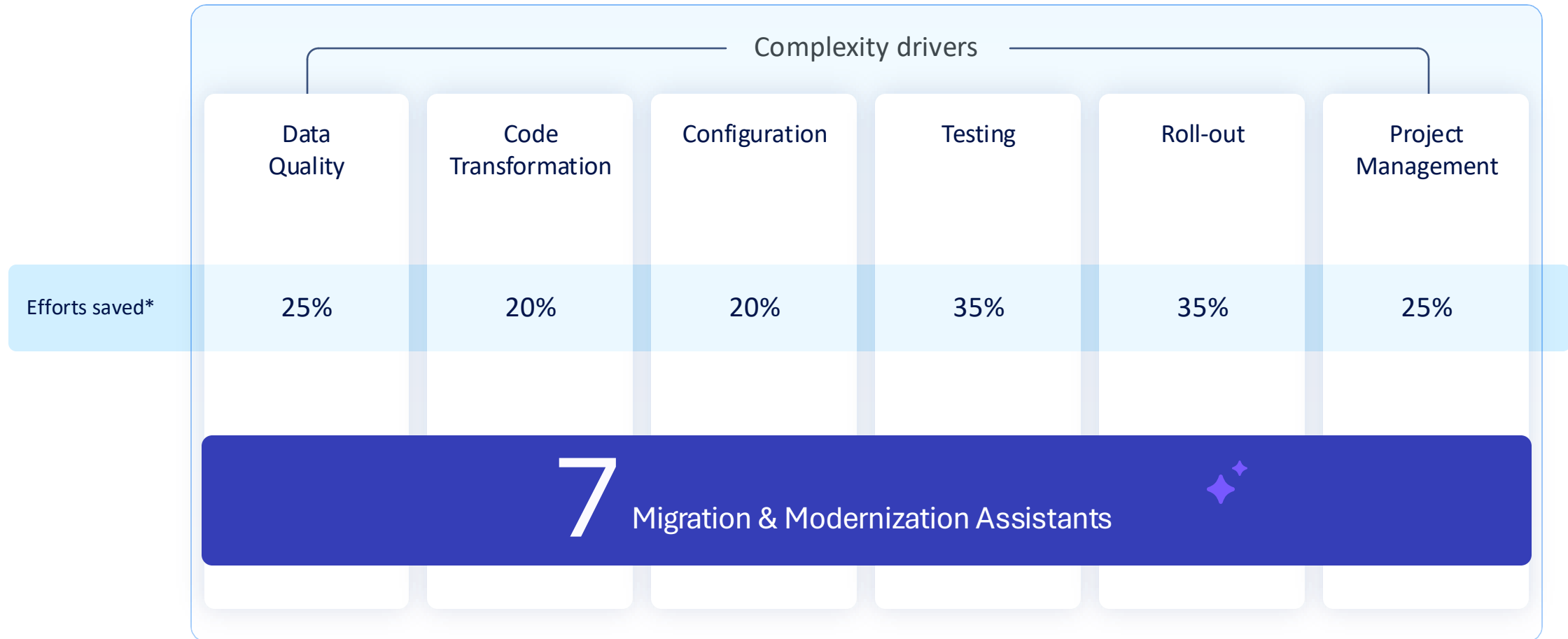
Providing AI capabilities and agents that reduce customer effort significantly when moving to Private Cloud via RISE, packaged and commercialized as one platform

## Step 2

### Long-shot: Nearly automated migration

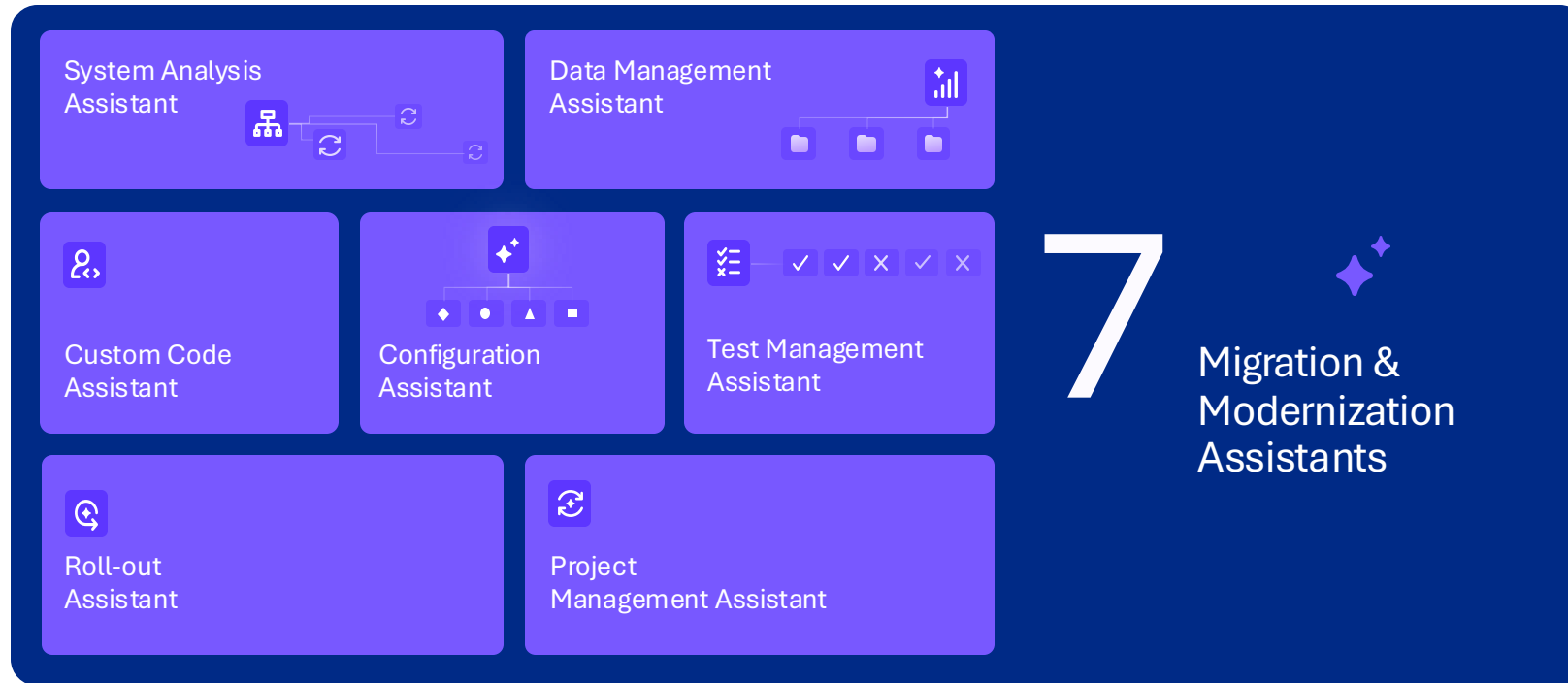
Build an autonomous migration fabric: Intelligent agents that continuously analyze legacy estates – data models, customizations, and extensions – then automatically design, configure, test, and execute end-to-end migrations to the cloud solutions with minimal human touch

# Migration effort is driven by six complexity drivers

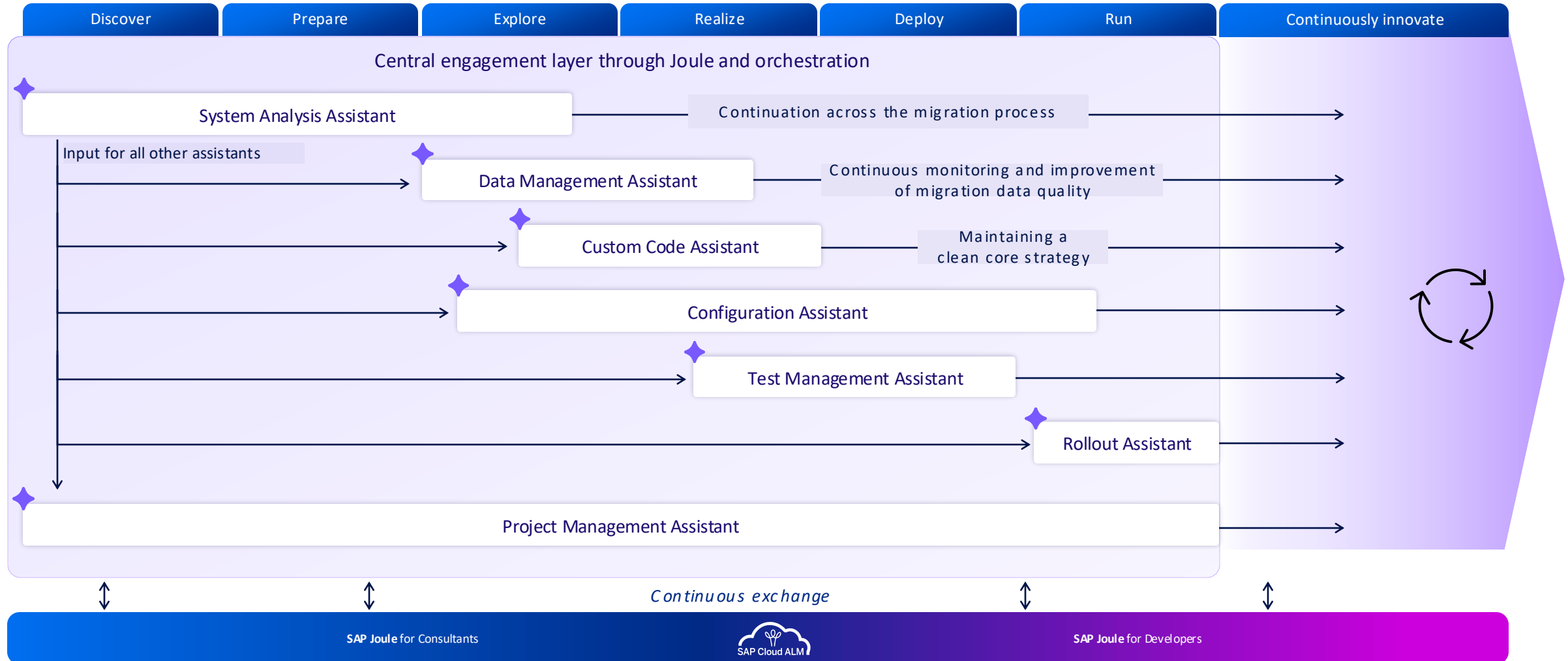


\*Effort reduction within the activities addressed by AI cases (e.g. 35% effort savings in test automation refers to all testing activities)

# Introducing seven migration and modernization assistants



# Assistants supporting ongoing modernization ✨

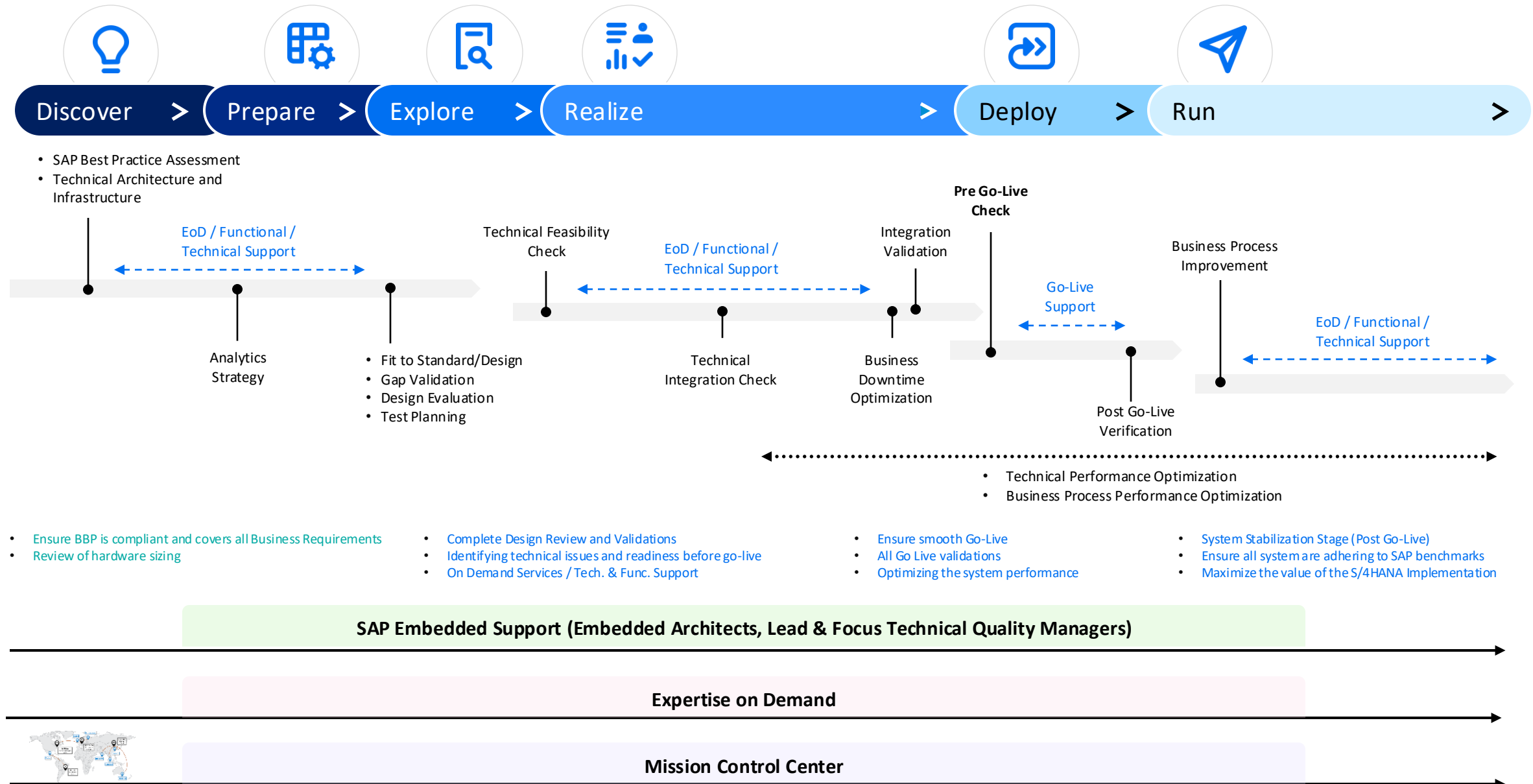


# SAP Services





# SAP Services for S/4HANA conversions & RISE migration





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